

```

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload satellite image
print("Upload a satellite image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Simulate Low Spatial Resolution
# -----

low_res = cv2.resize(
    img,
    (img.shape[1]//6, img.shape[0]//6),
    interpolation=cv2.INTER_AREA
)

low_res = cv2.resize(
    low_res,
    (img.shape[1], img.shape[0]),
    interpolation=cv2.INTER_NEAREST
)

# -----
# Enhance Contrast
# -----

lab = cv2.cvtColor(img, cv2.COLOR_RGB2LAB)
l, a, b = cv2.split(lab)

clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
l = clahe.apply(l)

enhanced = cv2.merge((l, a, b))
enhanced = cv2.cvtColor(enhanced, cv2.COLOR_LAB2RGB)

# -----
# Display
# -----

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Satellite Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(low_res)
plt.title("Low Resolution Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")

plt.show()

```

```
# -----  
# Explanation  
# -----  
  
print("Digital Image Fundamentals")  
print("- Higher sampling captures finer land details.")  
print("- Higher quantization improves intensity differences.")  
print("- Better resolution improves land classification.")  
  
print("\nImage Acquisition")  
print("- High-quality satellite sensors reduce noise.")  
print("- Proper lighting improves visibility.")  
print("- Good acquisition increases analysis accuracy.")  
  
print("\nApplications")  
print("- Agriculture")  
print("- Urban Planning")  
print("- Forest Monitoring")  
print("- Disaster Management")
```

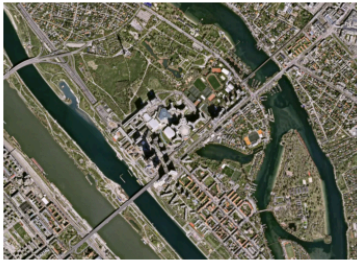
Upload a satellite image

Earth_from..._divide.webp

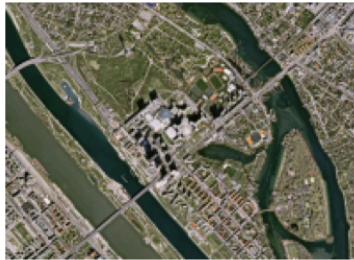
Earth_from_Space_Vienna_s_Danube_divide.webp(image/webp) - 303066 bytes, last modified: 14/07/2026 - 100% done

Saving Earth_from_Space_Vienna_s_Danube_divide.webp to Earth_from_Space_Vienna_s_Danube_divide.webp

Original Satellite Image



Low Resolution Image



Enhanced Image



Digital Image Fundamentals

- Higher sampling captures finer land details.
- Higher quantization improves intensity differences.
- Better resolution improves land classification.

Image Acquisition

- High-quality satellite sensors reduce noise.
- Proper lighting improves visibility.
- Good acquisition increases analysis accuracy.

Applications

- Agriculture
- Urban Planning
- Forest Monitoring
- Disaster Management

